

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO2: Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analyzing images.. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5 Total Marks:50 Annexure A

Analytical Problem Solving

Code of Conduct:

I CHEREDLA JASVINA/192324026 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Numerical Problems						
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	

Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	
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Annexure A

Analytical Problem Solving

Question 1: Salt-and-Pepper Noise Removal Before Edge Detection Scenario: A grayscale image of size 1024×1024 is captured using a surveillance camera under poor lighting conditions. The image is corrupted by salt-and-pepper noise with a noise density of 0.05. This means that some pixels are randomly changed to either black or white intensity values. The image has to be preprocessed before applying an edge detection algorithm. Since edge detection is sensitive to abrupt intensity variations, the selected filter should remove noise effectively while preserving important edge details.

Answer :

Salt-and-Pepper Noise Removal Before Edge Detection :

Given:

- Image size = 1024×1024
- Noise type = Salt-and-Pepper Noise
- Noise density = 0.05 (5%)
- Filter size = 3×3 Median Filter

Total number of pixels:

$$1024 \times 1024 = 1,048,576 \text{ pixels}$$

- a. Identify the most suitable filtering technique for removing salt-and-pepper noise without causing significant edge blurring.**

Answer:

The most suitable filtering technique is the Median Filter (3×3).

Reason:

- It effectively removes salt-and-pepper (impulse) noise.
- It preserves image edges better than linear filters.
- It replaces the noisy pixel with the median value of its neighboring pixels instead of averaging them.

- b. Explain why the selected filter performs better than an averaging filter for impulse noise removal.**

Answer:

Median Filter

- Removes isolated black and white noisy pixels.
- Uses the median value.
- Preserves edges and object boundaries.

Averaging (Mean) Filter

- Calculates the average of neighboring pixels.
- Salt-and-pepper noise contains extreme values (0 or 255).
- These extreme values influence the average, causing image blurring.

Median filtering performs better because:

- It is resistant to extreme intensity values.
- It removes impulse noise without blurring important edges.
- It maintains image details required for edge detection.

- c. **If a 3×3 median filter is applied, estimate the probability that a noisy center pixel will be corrected. Clearly state your assumptions.**

Answer:

Assumptions

- Noise occurs independently.
- Noise density =

$$p = 0.05$$

- A noisy center pixel is corrected if at least five pixels in the 3×3 window are noise-free.

There are 9 pixels in the window.

Probability of a pixel being noise-free:

$$1 - p = 0.95$$

Expected number of noise-free pixels:

$$9 \times 0.95 = 8.55$$

Since the probability that fewer than five pixels are clean is extremely small, the median is almost always determined by clean pixels.

Estimated probability

$$\approx 99\%$$

(or approximately 0.99)

Answer: A noisy center pixel has approximately a 99% probability of being corrected by the 3×3 median filter.

- d. **Estimate the total number of noisy pixels present in the original image.**

Answer:

Total pixels:

$$1024 \times 1024 = 1,048,576$$

Noise density:

$$0.05$$

Noisy pixels:

$$\begin{aligned} 1,048,576 \times 0.05 \\ = 52,428.8 \end{aligned}$$

Approximately,

$$52,429 \text{ noisy pixels}$$

- e. **Estimate the expected number of noisy pixels remaining after median filtering.**

Answer:

Assume:

- Median filter corrects 99% of noisy pixels.

Remaining noisy pixels:

$$\begin{aligned} 52,429 \times (1 - 0.99) \\ = 52,429 \times 0.01 \\ = 524.29 \end{aligned}$$

Approximately,

524 noisy pixels remain

f. Explain how the preprocessing strategy would change if the same image was affected by Gaussian noise instead of salt-and-pepper noise.

Answer:

Reason

Gaussian noise affects **all pixels slightly**, unlike salt-and-pepper noise, which changes only a few pixels to extreme values.

Suitable filters

- Gaussian Filter
- Wiener Filter
- Bilateral Filter (better edge preservation)

Why not Median Filter?

Median filtering is mainly designed for impulse noise and is less effective for Gaussian noise.

Therefore

For Gaussian noise:

- Use a **Gaussian filter** before edge detection.
- It smooths random intensity variations while reducing noise.
- Edge preservation can be improved using bilateral or Wiener filtering.

Question 2: Discretization of a Continuous Image

Scenario : An analog grayscale image has a smooth sinusoidal intensity variation along the horizontal direction. The intensity changes continuously with respect to the spatial coordinate x , while it remains constant along the y -direction. The image is defined over a spatial range of 0 to 255 in both directions. This continuous image has to be converted into a digital image for storage and further processing in a computer vision system.

Answers :

Given:

- Image is analog (continuous).
- Intensity varies as a smooth sinusoidal wave along the x -direction.
- Intensity is constant along the y -direction.
- Spatial range = 0 to 255 in both x and y directions.

a. Explain how the continuous image can be converted into a digital image using sampling and quantization.

A continuous image is converted into a digital image in two steps:

1. Sampling

- Sampling converts the continuous spatial coordinates into discrete pixels.
- The image is sampled at regular intervals along the x and y directions.
- This determines the image resolution.

2. Quantization

- Quantization converts the continuous intensity values into a finite number of gray levels.
- For an 8-bit image, each pixel intensity is represented using 256 gray levels (0–255).

Answer:

The analog image is digitized by first sampling the continuous image into pixels and then quantizing the pixel intensities into discrete gray levels (0–255).

b. Identify the spatial variation pattern of the image in the x-direction.

The image intensity varies as a smooth sinusoidal wave along the x-direction.

Therefore,

- Spatial variation = Sinusoidal (Periodic)
- Along the y-direction, the intensity remains constant.

Answer:

The spatial variation along the x-direction is sinusoidal (periodic).

c. Determine the minimum sampling requirement needed to avoid aliasing in the x direction.

According to the Nyquist Sampling Theorem:

The sampling frequency must be at least twice the highest spatial frequency present in the image.

If the highest spatial frequency is ,

$$f_s \geq 2f_{\max}$$

Therefore,

The minimum sampling rate required is:

$$\boxed{f_s \geq 2f_{\max}}$$

This is called the Nyquist rate.

d. Explain what will happen if the image is sampled below the required sampling rate.

If the sampling rate is less than the Nyquist rate:

- Aliasing occurs.
- High-frequency details appear as low-frequency patterns.
- False patterns and distortions appear.
- Fine image details are lost.
- The reconstructed image becomes inaccurate.

Answer:

Sampling below the Nyquist rate causes aliasing, leading to distortion, loss of detail, and incorrect image representation.

e. Suggest a suitable 8-bit quantization scheme for storing the image.

An 8-bit uniform quantization scheme is suitable.

- **Number of bits** = 8
- **Number of gray levels:**

$$2^8 = 256$$

Intensity values range from:

$$0 \text{ to } 255$$

Each pixel is stored using 8 bits (1 byte).

Answer:

Use 8-bit uniform quantization with 256 gray levels (0–255).

f. Discuss the effect of quantization on image quality, storage size, and intensity accuracy.

Image Quality

- Higher bit depth gives smoother intensity transitions.
- Lower bit depth causes visible banding and loss of detail.

Storage Size

- More bits per pixel require more storage space.
- 8-bit images require 1 byte per pixel, providing a good balance between quality and storage.

Intensity Accuracy

- Higher bit depth represents the original intensities more accurately.
- Lower bit depth introduces quantization error because many intensity values are mapped to fewer gray levels.

Answer:

8-bit quantization provides 256 gray levels, offering good image quality with moderate storage requirements. Increasing the bit depth improves intensity accuracy but increases storage size, while reducing the bit depth saves storage but introduces quantization errors and reduces image quality.

Question 3: Edge Detection in Thin Line Images

Scenario: A 512×512 grayscale image contains very thin vertical and horizontal white lines of 1-pixel thickness on a dark uniform background. Such images are commonly used in document analysis, engineering drawing interpretation, circuit diagram processing, and road-lane detection.

The objective is to detect the line boundaries accurately without losing thin edge information.

Answers :

Given:

- Image size = 512×512
- Thin vertical and horizontal white lines
- Line thickness = 1 pixel
- Background = Dark
- Objective = Detect thin edges accurately

Compare Sobel, Prewitt, and Canny edge detectors for detecting 1-pixel-thick vertical and horizontal lines.

Edge Detector	Advantages	Disadvantages	Suitability
Sobel	Detects edges using gradient, provides some noise smoothing	Produces thicker edges and may miss very thin lines	Good
Prewitt	Simple and computationally efficient	More sensitive to noise than Sobel, less accurate	Fair
Canny	High detection accuracy, detects weak edges, good localization, uses non-maximum suppression	More computationally expensive	Best choice

Answer:

For detecting 1-pixel-thick vertical and horizontal lines, the Canny edge detector is the most suitable because it provides accurate edge localization, preserves thin edges, and reduces false detections.

b. For a horizontal edge located near pixel position (256, 256), compute the Sobel gradient response using the standard Sobel operator.

Standard Sobel Kernels

Horizontal gradient (G_x):

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

Vertical gradient (G_y):

$$G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

For a horizontal edge, the intensity changes vertically.

Therefore,

- **Horizontal gradient:**

$$G_x \approx 0$$

- **Vertical gradient:**

$$G_y = \text{Maximum}$$

Gradient magnitude:

$$G = \sqrt{G_x^2 + G_y^2}$$

Since $G_x = 0$,

$$G = |G_y|$$

Gradient direction:

$$\theta = \tan^{-1}\left(\frac{G_y}{G_x}\right) = \tan^{-1}(\infty) = 90^\circ$$

Answer:

- $G_x \approx 0$
- $G_y = \text{Maximum}$
- Gradient magnitude = $|G_y|$
- Gradient direction = 90°

c. Explain how the gradient magnitude and gradient direction help in identifying the strength and orientation of an edge.

Gradient Magnitude

$$G = \sqrt{G_x^2 + G_y^2}$$

- Indicates the strength of an edge.
- Larger magnitude means a stronger edge.
- Smaller magnitude indicates a weak edge or noise.

Gradient Direction

$$\theta = \tan^{-1}\left(\frac{G_y}{G_x}\right)$$

- Indicates the orientation of the edge.
- Helps determine whether the edge is horizontal, vertical, or diagonal.
- Used in non-maximum suppression in the Canny detector.

d. Discuss the role of Gaussian smoothing in the Canny edge detection process.

Purpose

Gaussian smoothing is the first step in the Canny algorithm.

- Removes random noise.

- Reduces small intensity variations.
- Prevents false edge detection.
- Produces smoother gradients before edge detection.

Answer:

Gaussian smoothing reduces image noise before calculating gradients, improving the accuracy and reliability of edge detection.

e. Explain how Gaussian smoothing may affect edge localization, especially for thin lines.

Although Gaussian smoothing removes noise:

- It slightly blurs the image.
- Very thin (1-pixel) lines may become wider or weaker.
- Exact edge position may shift slightly.
- Excessive smoothing can remove fine details.

Answer:

Gaussian smoothing improves noise removal but may reduce edge localization accuracy by blurring very thin lines and slightly shifting their detected positions.

f. If the Canny threshold is reduced to half of its original value, explain the expected changes in detected edges, weak edges, and false edges.

Reducing the threshold makes the detector more sensitive.

Effects

Detected edges

- More edges are detected.

Weak edges

- More weak edges are retained.

False edges

- More noise may be detected as edges, increasing false edges.

Question 4: Feature Extraction for Handwritten Digit Recognition Scenario: A computer vision system is being developed for recognizing handwritten digits from scanned images. The handwritten digits vary in size, orientation, stroke thickness, and writing style. Therefore, robust feature extraction is required before classification.

Answers :

- Recognize handwritten digits from scanned images.
- **Digits vary in:**
 - Size
 - Orientation
 - Stroke thickness
 - Writing style
- Robust feature extraction is required before classification.

a. Explain the working principles of SIFT, SURF, and ORB feature extraction methods.

1. SIFT (Scale-Invariant Feature Transform)

Working Principle:

- Detects distinctive keypoints using Difference of Gaussian (DoG).
- Assigns an orientation to each keypoint based on local gradients.
- Creates a descriptor from the surrounding gradient information.

Features:

- Scale invariant
- Rotation invariant
- Highly accurate
- Robust to illumination changes

2. SURF (Speeded-Up Robust Features)**Working Principle:**

- Detects keypoints using the Hessian Matrix.
- Uses integral images for faster computation.
- Generates descriptors based on Haar wavelet responses.

Features:

- Faster than SIFT
- Scale invariant
- Rotation invariant
- Good accuracy

3. ORB (Oriented FAST and Rotated BRIEF)**Working Principle:**

- Detects keypoints using the FAST algorithm.
- Assigns orientation using the intensity centroid method.
- Describes features using the BRIEF descriptor.

Features:

- Very fast
- Rotation invariant
- Efficient for real-time applications
- Lower computational cost

b. Compare SIFT, SURF, and ORB in terms of scale invariance, rotation invariance, computational cost, and real-time suitability.

- SIFT (Scale-Invariant Feature Transform): SIFT is scale invariant and rotation invariant. It provides highly accurate feature matching but has a high computational cost, making it less suitable for real-time applications.
- SURF (Speeded-Up Robust Features): SURF is also scale invariant and rotation invariant. It is faster than SIFT because it uses integral images and has a moderate computational cost. It is reasonably suitable for real-time applications.
- ORB (Oriented FAST and Rotated BRIEF): ORB is rotation invariant and partially scale invariant. It has a low computational cost and is much faster than both SIFT and SURF, making it the best choice for real-time applications.

Conclusion:

SIFT provides the highest accuracy but is slow. SURF offers a balance between speed and accuracy. ORB is the fastest and most suitable for real-time handwritten digit recognition due to its low computational cost.

c. If a digit image is rotated by 45 degrees, identify which feature extraction methods can still produce reliable matching features.

A rotation of 45° changes the orientation of the image.

Since SIFT, SURF, and ORB are all rotation invariant, they can still detect and match features correctly.

Answer:

- ✓ SIFT
- ✓ SURF
- ✓ ORB

All three methods can produce reliable matching features for a 45° rotated image.

d. If the average number of keypoints detected per image is 120 and 50 images are processed, compute the total number of keypoints detected.

Average keypoints per image:

120

Number of images:

50

Total keypoints:

$$120 \times 50 = 6000$$

Answer:

6000 keypoints

e. Suggest a suitable dimensionality reduction technique to reduce the feature size while retaining 95% of the useful information.

The most suitable technique is Principal Component Analysis (PCA).

Why PCA?

- Reduces the number of features.
- Retains about 95% of the variance (useful information).
- Removes redundant and correlated features.
- Improves computational efficiency.

Answer:

Use Principal Component Analysis (PCA) while selecting enough principal components to preserve 95% of the variance.

f. Explain how dimensionality reduction improves classification efficiency in handwritten digit recognition.

Dimensionality reduction provides several benefits:

- Reduces the number of input features.
- Decreases memory usage.
- Speeds up training and testing.
- Removes redundant and noisy features.
- Reduces overfitting.
- Improves classifier performance and accuracy.

Answer:

Dimensionality reduction improves handwritten digit recognition by reducing feature size, lowering computational cost, minimizing noise and redundancy, reducing overfitting, and enabling faster and more accurate classification.

Question 5: Morphological Preprocessing for Character Recognition Scenario: A binary image of size 256×256 contains scanned handwritten characters. Due to poor scanning quality, the image contains small white noise dots in the background and small black gaps inside the character strokes.

Before applying character recognition, the image must be cleaned using morphological preprocessing techniques.

Answers :

Given:

- Binary image size = 256×256
- Contains handwritten characters
- **Problems:**
 - Small white noise dots in the background
 - Small black gaps/holes inside character strokes
- **Goal:** Clean the image before character recognition

a. Identify the morphological operation suitable for removing small white noise dots from the background.

The suitable morphological operation is Opening.

Reason:

- Opening = Erosion followed by Dilation.
- It removes small white noise dots while preserving the main shape of the characters.

Answer:

Opening is used to remove small white noise dots from the background.

b. Identify the morphological operation suitable for filling small gaps or holes inside the character strokes.

The suitable morphological operation is Closing.

Reason:

- Closing = Dilation followed by Erosion.
- It fills small holes and gaps inside the character strokes and connects broken parts.

Answer:

Closing is used to fill small gaps or holes inside the character strokes.

c. Explain the difference between opening and closing operations using simple image processing terminology.

Opening:

- Performs erosion followed by dilation.
- Removes small white objects or noise.
- Smooths object boundaries without significantly changing the object size.

Closing:

- Performs dilation followed by erosion.
- Fills small holes and gaps.
- Connects nearby object parts and smooths boundaries.

Difference:

Opening removes small unwanted white objects, whereas closing fills small holes and connects broken parts of the objects.

d. If a 3×3 square structuring element is used, explain how it affects thin character strokes.

A 3×3 square structuring element has a moderate effect:

- Removes small noise effectively.
- Fills tiny gaps in strokes.
- Slightly thins or smooths very thin character strokes.
- Usually preserves the overall character shape if applied carefully.

Answer:

A 3×3 structuring element removes small noise and fills tiny gaps while causing only slight changes to thin character strokes.

e. Discuss the risk of using a large structuring element during preprocessing.

Using a large structuring element may:

- Remove important character details.
- Distort the shape of characters.
- Merge nearby characters together.
- Break thin strokes during opening.
- Overfill gaps during closing.
- Reduce recognition accuracy.

Answer:

A large structuring element may remove useful details, distort character shapes, merge adjacent characters, and decrease the accuracy of character recognition.

f. Propose a complete preprocessing pipeline for handwritten character recognition, starting from image acquisition to feature extraction.

A suitable preprocessing pipeline is:

1. Image Acquisition – Capture or scan the handwritten document.
2. Grayscale Conversion – Convert the image to grayscale (if needed).
3. Noise Removal – Remove noise using median filtering or morphological operations.
4. Binarization – Convert the grayscale image into a binary image using thresholding.
5. Morphological Processing – Apply opening to remove noise and closing to fill gaps.
6. Segmentation – Separate individual characters.
7. Normalization – Resize and align characters to a standard size.
8. Feature Extraction – Extract features using methods such as SIFT, SURF, ORB, or HOG before classification.